DAVID NAVARRO GIRONÉS

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RESEARCH HIGHLIGHTS

Research interests: Cosmology, weak gravitational lensing, intrinsic alignments of galaxies, photometric redshift estimation, hydrodynamical simulations.
Publications: 25 refereed publications (3 first-author), 5 submitted manuscripts. 5 publications in preparation.
Talks: 5 invited, 10 contributed.
Software: Lead developer of IATheory. Developer of IACorr, cloelib, and CLOE.
Leadership: Lead of Euclid DR1 IA analysis. Supervisor/co-supervisor of 4 students (2 BSc, 1 MSc, 1 PhD).
Awards and Funding: Marie Skłodowska-Curie Actions Seal of Excellence. Spanish PhD fellowship.

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate — Since 01/11/2024
Leiden Observatory, Leiden University, Leiden, the Netherlands

PhD Candidate — 01/10/2020–15/10/2024
Institute of Space Sciences (ICE, IEEC-CSIC), Cerdanyola del Valles, Spain

Undergraduate Research Assistant — 10/09/2015–01/08/2018
Institut de Física d'Altes Energies (IFAE), Cerdanyola del Vallès, Spain

EDUCATION

PhD in Physics — 01/10/2020–15/10/2024
Thesis: A study of photometric redshifts and intrinsic alignments with PAUS and Euclid
Advisors: Dr. Martin Crocce and Dr. Enrique Gaztañaga
Institute of Space Sciences (ICE, IEEC-CSIC), Cerdanyola del Valles, Spain

Master's degree in Astrophysics, Cosmology and Particle Physics — 10/09/2018–10/09/2019
Universitat de Barcelona (UB), Barcelona, Spain

Bachelor's degree in Physics — 01/09/2013–01/09/2018
Universitat Autònoma de Barcelona (UAB), Cerdanyola del Valles, Spain

Bachelor's degree in Chemistry — 01/09/2013–01/09/2018
Universitat Autònoma de Barcelona (UAB), Cerdanyola del Valles, Spain

PUBLICATIONS

I have co-authored 30 published, accepted, or submitted articles (∼1000 citations; h-index: 8). The complete public record is available on NASA ADS. My most relevant publications are listed below.

• *Impact of galaxy intrinsic alignment modelling choices on Euclid 3×2pt cosmology.*
Euclid Collaboration: **D. Navarro-Gironés** et al.
Accepted in A&A, 2026 https://doi.org/10.1051/0004-6361/202659548

• *The PAU Survey: Measuring intrinsic alignment of galaxies for the PAUS wide fields.*
D. Navarro-Gironés et al.
MNRAS, Volume 545, January 2026 https://doi.org/10.1093/mnras/staf1630

• *The PAU Survey: Photometric redshift estimation for the PAUS wide fields.*
D. Navarro-Gironés et al.
MNRAS, Volume 534, October 2024 https://doi.org/10.1093/mnras/stae1686

• *The PAU Survey and Euclid: Analysing photometric redshifts from realistically simulated narrow-band photometry with Flagship 2.*
A. Wittje, H. Hildebrandt, **D. Navarro-Gironés** et al.
Submitted to A&A, 2026 https://arxiv.org/abs/2606.25812

• *Intrinsic alignments in the FLAMINGO simulations with two-point statistics.*
A. Herle, N. E. Chisari, H. Hoekstra, **D. Navarro-Gironés** et al.
A&A, Volume 711, A235, July 2026 https://doi.org/10.1051/0004-6361/202659078

Articles in preparation:

• *Constraining intrinsic-alignment amplitudes as a function of galaxy colour, redshift, and luminosity.*
Euclid Collaboration: **D. Navarro-Gironés** et al.

• *Exploring intrinsic galaxy alignments with marked correlation functions in the FLAMINGO simulations.*
Y. Shang, **D. Navarro-Gironés** et al.

• *Intrinsic galaxy alignments as a contaminant to redshift-space distortions with image simulations.*
N. Kaijser, **D. Navarro-Gironés** et al.

• *Tailoring intrinsic-alignment priors to galaxy demographics for next-generation weak lensing.*
J. McCullough, J. Siegel, A. Amon, B. Joachimi, D. Gruen, **D. Navarro-Gironés**.

• *Revisiting cross-correlations between gravitational waves and intrinsic alignments in Stage-IV surveys.*
Y. Chang, G. Cañas-Herrera, **D. Navarro-Gironés**, D. Rana.

COLLABORATION MEMBERSHIP

Physics of the Accelerating Universe Survey (PAUS) Since 01/10/2020
Builder status. Led two key analyses for PAUS science: estimation of photometric redshifts and measurement and modelling of intrinsic alignments of galaxies.

***Euclid* Consortium** Since 06/03/2023
Lead of the Euclid DR1 IA analysis. Lead author of Euclid forecasting paper on IA modelling. Contributor to cloelib/CLOE likelihood development. Contributor to Euclid 3×2pt cosmology.

ARRAKIHS Since 26/05/2025

LSST DESC Since 07/07/2025

FUNDING

Travel grant from Duke University – 600$ 05/08/2025
Attendance at the CAROLINA Workshop (echoIA) at Duke University, North Carolina.

Travel grant from LSST DESC – 2.700$ 16/06/2025
Attendance at the Forecasting + IA LSST-DESC Workshop at Durham, North Carolina.

Travel grant from the Leiden University Fund – 1.000€ 08/05/2025
Attendance at the Cosmic Cartography with Roman conference at Baltimore, Maryland.

PhD fellowship from the Spanish Ministry of Science and Innovation – 98.250€ 01/10/2020–01/10/2024
Ayudas para contratos predoctorales para la formación de doctores.

Erasmus grant from the European Union – 840€ 30/01/2023–10/02/2023
Grant supporting a two-week research stay at a European university or research centre.

University grant from the Spanish Ministry of Education – 10.000€ 01/09/2013–01/09/2018
Full scholarship for the Physics Bachelor and Chemistry Bachelor tuition

University grant from CatalunyaCaixa – 1.000€ 18/06/2013
Awarded to the 500 highest university entrance qualifications in Catalonia.

AWARDS AND HONOURS

Seal of excellence - Marie Skłodowska-Curie Actions 16/02/2026
Quality label awarded to applicants who achieved a high score in their applications.

Distincions PAU 01/09/2013
Awarded for excellence in the Catalan university entrance examinations.

SUPERVISING AND MENTORING ACTIVITIES

BSc summer student Yu-Ting Chang – Yale University 15/06/2026–21/08/2026
Interplay between intrinsic alignments of galaxies and gravitational waves.

Visiting PhD student Francisco Zarate – Universidad Nacional de San Juan Since 01/11/2025
Photometric redshift estimation on regions with high galactic extinction.

MSc thesis of Noah Kaijser – Leiden University Since 01/09/2025
Study of intrinsic alignments of galaxies as a contaminant to redshift-space distortions.

Visiting BSc student Yuxin Shang – Tsinghua University Since 11/07/2025
Exploring intrinsic galaxy alignments with marked correlation functions.

SCIENTIFIC SOFTWARE

IATheory – Lead developer: Package for IA modelling
IACorr – Developer: Package for measuring IA
cloelib – Developer: Euclid likelihood and inference code
CLOE – Developer: previous Euclid likelihood and inference code

INVITED TALKS

Prof. Marcelle Soares-Santos group seminar 05/05/2026
University of Zurich, Switzerland
Title: *Towards precision cosmology with Euclid: intrinsic alignments and photometric redshifts*

CAROLINA meeting 15/09/2025
Duke University, United States of America
Title: *Observations on IA (overview talk)*

Euclid Consortium Meeting 27/03/2025
Leiden, the Netherlands
Title: *Impact of IA in 3×2pt analyses in Euclid*

WAVES Team Meeting 2024 04/12/2024
Perth, Australia
Title: *Physics of the Accelerating Universe Survey overview*

Institute of Cosmology and Gravitation WL seminar 28/11/2024
University of Portsmouth, United Kingdom
Title: *Galaxy clustering and intrinsic alignments with the PAU Survey*

CONTRIBUTED TALKS

European Astronomical Society Meeting 29/06/2026
Lausanne, Switzerland
Title: *Impact of galaxy intrinsic alignments modelling choices on Euclid 3×2pt cosmology*

Euclid Spain meeting 23/02/2026
Tenerife, Spain
Title: *Impact of galaxy intrinsic alignments modelling choices on Euclid 3×2pt cosmology*

Cosmic Cartography with Roman 16/07/2025
Baltimore, United States of America
Title: *Measuring Intrinsic Alignments with the Physics of the Accelerating Universe Survey*

NOVA workshop 24/01/2025
Groningen, the Netherlands
Title: *Measuring IA with the PAU Survey*

echoIA LILAC workshop 09/05/2024
Center for Astrophysics | Harvard & Smithsonian, USA
Title: *Measuring IA with the PAU Survey*

IX Meeting on Fundamental Cosmology 23/11/2023
IACTEC (La Laguna, Canary Islands)
Title: *Photometric redshifts in the PAUS wide fields*

Inter Science Task Force-Non Linear (IST-NL) workshop 02/10/2023
ICE (Barcelona)
Title: *Implementation of the IA TATT model in IST-NL / CLOE*

Hol-IA Workshop 15/03/2023
Lorentz Center (Leiden)
Title: *Measuring IA with the PAU Survey*

2nd Roman Juszkiewicz Symposium 26/09/2022
Nicolaus Copernicus Astronomical Centre (Warsaw)
Title: *Measuring IA with the PAU Survey*

IberiCOS2022 04/05/2022
Institute of Space Sciences
Title: *Measuring IA with the PAU Survey*

RESEARCH STAYS

IRAP-OMP (Toulouse) 30/01/2023–10/02/2023
Two-week stay with Dr. Isaac Tutusaus to start implementing an intrinsic alignment model (TATT) module in the likelihood code of Euclid (CLOE).

Ruhr-Universität Bochum 28/03/2022–31/05/2022
Two-month stay in Prof. Hendrik Hildebrandt's group to work on a PAUS collaboration project related to photometric redshift estimation and intrinsic alignment measurements.

SERVICE AND LEADERSHIP

Refereeing:

- Referee for The Astrophysical Journal — Since 03/08/2026
- Internal referee for the Euclid Collaboration — Since 12/12/2025

Workshop and Conference Organisation:

- LOC UNBIAS: echoIA Workshop — 31/08/2026–04/09/2026
- LOC COSMO-26 — 24/08/2026–28/08/2026
- SOC European Astronomical Society Annual Meeting: Euclid Symposium — 29/06/2026–03/07/2026
- SOC Euclid Annual Meeting — 25/05/2026–29/05/2026
- LOC BCN-MAD Meeting — 17/04/2024

Community, Inclusivity and Coordination:

- Postdoctoral Ambassador Leiden Observatory — Since 20/09/2024
- Euclid Consortium Diversity Committee — Since 10/05/2024
- PAUS Data Release coordination — 18/09/2024
- ICE Equity Committee — 01/09/2021–01/09/2022